EXPERIMENT-2

ROLL NO: 240701244 NAME: KAVITHA R

Design and Implementation of CLI, GUI, and Voice User Interfaces (VUI) for an Intelligent Expense Tracking System Using Python

Aim:

The aim of this project is to design, develop, and compare three different user interfaces, namely Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI), for performing the same task. The project also focuses on evaluating user satisfaction and usability of each interface using Python, with Tkinter for GUI development, Speech Recognition for VUI implementation, and Terminal for CLI interaction.

PROCEDURE:

CLI (Command Line Interface):

CLI implementation where users can add, view, and remove tasks using the terminal.

CODE:

```
expenses = []
def add_expense():
amount = float(input("Enter amount: "))
reason = input("Enter reason: ")
expenses.append((amount, reason))
print("Expense added successfully!")
def view_expenses():
if expenses:
print("\nYour Expenses:")
    for i, (amount, reason) in enumerate(expenses, 1):
      print(f"{i}. Rs.{amount} - {reason}")
```

```
else:
    print("No expenses found.")
def total_spent():
total = sum(amount for amount, _ in expenses)
print(f"\nTotal Money Spent: Rs.{total}")
def delete_expense():
view_expenses()
if expenses:
num = int(input("Enter expense number to delete: "))
if 0 < num <= len(expenses):
removed = expenses.pop(num - 1)
print(f"Deleted: Rs.{removed[0]} - {removed[1]}")
else:
 print("Invalid number.")
def main():
  while True:
    print("\n------ Expense Tracker ------")
    print("1. Add Expense")
    print("2. View Expenses")
    print("3. View Total Spent")
    print("4. Delete Expense")
    print("5. Exit")
    choice = input("Enter your choice: ")
    if choice == '1':
      add_expense()
    elif choice == '2':
      view_expenses()
    elif choice == '3':
      total_spent()
    elif choice == '4':
      delete_expense()
```

```
        elif choice == '5':
            print("Exiting... Goodbye!")
            break
        else:
            print("Invalid choice! Try again.")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
------ Expense Tracker ------
1. Add Expense
2. View Expenses
3. View Total Spent
4. Delete Expense
5. Exit
Enter your choice: 1
Enter amount: 100
Enter reason: Book
Expense added successfully!

------ Expense Tracker ------
1. Add Expense
2. View Expenses
3. View Total Spent
4. Delete Expense
5. Exit
Enter your choice: 2
Enter your choice: 2

Your Expenses:
Your Expenses:
1. Rs.100.0 - Book

------ Expense Tracker ------
1. Add Expense
2. View Expenses
3. View Total Spent
4. Delete Expense
5. Exit
Enter your choice: 3

Total Money Spent: Rs.100.0
```

```
------ Expense Tracker ------
1. Add Expense
2. View Expenses
3. View Total Spent
4. Delete Expense
5. Exit
Enter your choice: 4

Your Expenses:
1. Rs.100.0 - Book
Enter expense number to delete: 1
Deleted: Rs.100.0 - Book

------ Expense Tracker ------
1. Add Expense
2. View Expenses
3. View Total Spent
4. Delete Expense
5. Exit
Enter your choice: 5
Exiting... Goodbye!
```

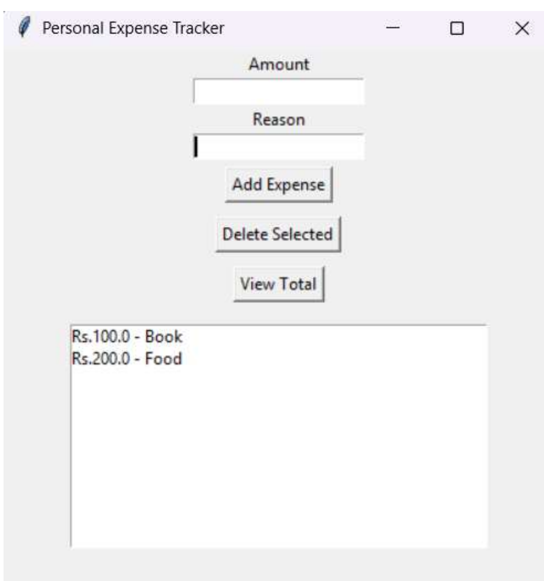
GUI (Graphical User Interface):

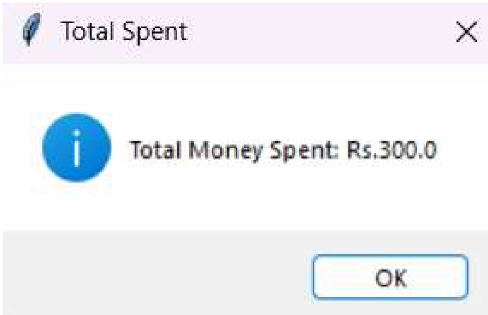
CODE:

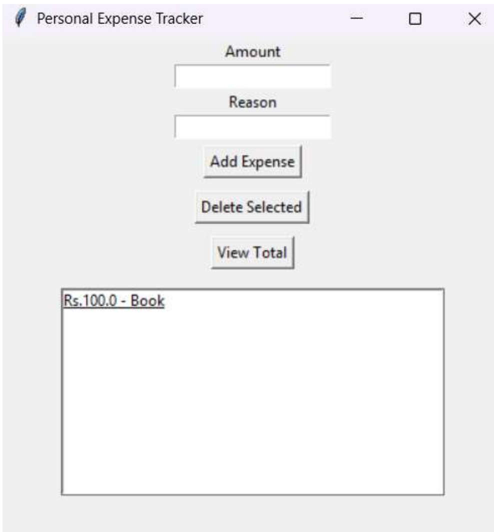
```
import tkinter as tk
from tkinter import messagebox
expenses = []
def add_expense():
  amount = amount_entry.get()
  reason = reason_entry.get()
  if amount == "" or reason == "":
    messagebox.showwarning("Input Error", "Please enter both amount and reason")
    return
  try:
    amount = float(amount)
  except:
    messagebox.showerror("Invalid Input", "Amount must be a number")
    return
  expenses.append((amount, reason))
  listbox.insert(tk.END, f"Rs.{amount} - {reason}")
  amount_entry.delete(0, tk.END)
  reason_entry.delete(0, tk.END)
def delete_expense():
  selected = listbox.curselection()
  if not selected:
    messagebox.showwarning("Selection Error", "Select an item to delete")
    return
  index = selected[0]
  listbox.delete(index)
  expenses.pop(index)
def total_spent():
```

```
    total = sum(amount for amount, _ in expenses)
    messagebox.showinfo("Total Spent", f"Total Money Spent: Rs.{total}")
root = tk.Tk()
root.title("Personal Expense Tracker")
root.geometry("400x400")
tk.Label(root, text="Amount").pack()
amount_entry = tk.Entry(root)
amount_entry.pack()
tk.Label(root, text="Reason").pack()
reason_entry = tk.Entry(root)
tk.Button(root, text="Add Expense", command=add_expense).pack(pady=5)
tk.Button(root, text="Delete Selected", command=delete_expense).pack(pady=5)
tk.Button(root, text="View Total", command=total_spent).pack(pady=5)
listbox = tk.Listbox(root, width=50)
listbox.pack(pady=10)
root.mainloop()
```

OUTPUT:







VUI (Voice User Interface)

```
import speech_recognition as sr
import pyttsx3
engine = pyttsx3.init()
voices = engine.getProperty('voices')
engine.setProperty('voice', voices[1].id)
engine.setProperty('rate', 170)
expenses = []
BOT_NAME = "Ava"
def speak(text):
  print(f"{BOT_NAME}:", text)
  engine.say(text)
  engine.runAndWait()
def listen():
```

```python
    r = sr.Recognizer()
    r.energy_threshold = 300
    r.pause_threshold = 0.8
    with sr.Microphone() as source:
        print("Listening... Speak clearly")
        audio = r.listen(source, timeout=5, phrase_time_limit=5)
    try:
        command = r.recognize_google(audio, language="en-IN")
        print("You said:", command)
        return command.lower()
    except:
        return ""
def add_expense():
    speak("Please say the amount")
    amount = input("Amount: ")
    try:
        amount = float(amount)
    except:
        speak("Invalid amount")
        return
    speak("Please say the reason")
    reason = input("Reason: ")
    expenses.append((amount, reason))
    speak("Expense added successfully")
def view_expense():
    if not expenses:
        speak("No expenses found")
        return
    speak("Here are your expenses")
    for i, (amt, reason) in enumerate(expenses, start=1):
        print(f"{i}. {amt} rupees - {reason}")
```

```
        speak(f"{amt} rupees for {reason}")
def delete_expense():
    if not expenses:
        speak("No expenses to delete")
        return
    view_expense()
    speak("Enter the expense number to delete")
    try:
        index = int(input("Enter number: "))
        removed = expenses.pop(index - 1)
        speak(f"Deleted {removed[0]} rupees for {removed[1]}")
    except:
        speak("Invalid selection")
def total_spent():
    total = sum(amt for amt, _ in expenses)
    speak(f"Total spent is {total} rupees")
def main():
    speak("Hello! I am Ava, your smart expense assistant")
    while True:
        speak("Say add, view, delete, total, or exit")
        command = listen()
        if command == "":
            speak("I did not hear you, please repeat")
            continue
        if "add" in command:
            add_expense()
        elif "view" in command or "show" in command or "see" in command:
            view_expense()
        elif "delete" in command or "remove" in command:
            delete_expense()
        elif "total" in command or "sum" in command:
```

```
            total_spent()
        elif "exit" in command or "quit" in command or "stop" in command:
            speak("Goodbye! Have a nice day")
            break
        else:
            speak("Command not recognized. Please say add, view, delete, total, or exit")

if __name__ == "__main__":
    main()
```

OUTPUT:

```
PS C:\Users\kavit\Downloads\UID> py -3.10 expense_tracker_vui.py
Ava: Hello! I am Ava, your smart expense assistant
Ava: Say add expense, view expense, delete expense, total spent, or exit
Listening...
You said: add expense
Ava: Please say the amount
Amount: 200
Ava: Please say the reason
Reason: Book
Ava: Expense added successfully
```

```
Ava: Say add expense, view expense, delete expense, total spent, or exit
Listening...
You said: view expense
Ava: Here are your expenses
1. 200.0 rupees - Book
Ava: 200.0 rupees for Book
2. 30.0 rupees - pen
Ava: 30.0 rupees for pen
```

```
Ava: Say add expense, view expense, delete expense, total spent, or exit
Listening...
You said: delete expense
Ava: Here are your expenses
1. 200.0 rupees - Book
Ava: 200.0 rupees for Book
2. 30.0 rupees - pen
Ava: 30.0 rupees for pen
Ava: Enter the expense number to delete
Enter number: 2
Ava: Deleted 30.0 rupees for pen
```

```
Ava: Say add expense, view expense, delete expense, total spent, or exit
Listening...
You said: total spent
Ava: Total spent is 200.0 rupees
Ava: Say add expense, view expense, delete expense, total spent, or exit
Listening...
You said: exit
Ava: Goodbye! Have a nice day
```